

4 Fair Division

4.1 Divisor Chooser Continued

Example 4.1.1 Suppose that Betty and Cade are sharing a cake which is half strawberry and half vanilla. If there are six slices of cake total, and cade values straberry 3 times as much as he values vanilla, how much is each slice of straberry worth to him?

Example 4.1.2 Your turn! If Betty (in the same problem as above) values strawberry 2 times as much as vanilla, how much is each slice of strawberry worth to betty?

Example 4.1.3 Troy and Abed are sharing a Subway sandwich, which is a foot long and half veggie and half chicken. Troy likes chicken 2 times as much as veggies, and Abed likes chicken and veggies equally. If Troy is the divider, describe how he will split the sandwich (only vertical slices are allowed). Which part will Abed choose?

Example 4.1.4 Your turn! Annie and Jeff are now going to share a sandwich, which is half veggie and half chicken, and is a foot long. Annie is a vegetarian, while Jeff likes veggie 3 times as much as he likes chicken. If Jeff is the divider, describe how he will split the sandwich (only vertical slices are allowed). Which part will Annie choose?

Example 4.1.5 Two retirees, Pierce and Buzz, purchase a vacation home together in Florida, with the agreement that they will split up the year into two pieces which will determine when each can use the home. Pierce loves Florida in the winter months, and dislikes summer, while Buzz likes all parts of the year equally. If Pierce is the divider, describe how he might divide up the year into two pieces. Which piece would Buzz choose? (e.g. Pierce could do something like the following: May-August is one piece, and the other is September-April). There are many correct answers to this, explain why yours produces a fair division.